MCQ A · integral, area and mean value

After S63; we always assume $a < b$.

For each question, only one option is correct. Just consult the solutions.

1. If $F' = f$ on $[a;b]$, then $\int_a^b f(x)\,\mathrm{d}x$ is:

A. $F(a) + F(b)$ B. $F(b) - F(a)$

C. $f(b) - f(a)$ D. 0

2. The integral of a negative function on $[a;b]$ is:

A. positive B. zero

C. negative or zero D. always a positive area

3. The average value of f on $[a;b]$ is:

A. $\frac{1}{b-a}\int_a^b f$ B. $\int_a^b f$

C. $f(a) + f(b)$ D. $b - a$

4. The geometric area between the curve and the x-axis requires:

A. preserving the signs B. split where the sign changes

C. a second derivative D. a random draw

5. An antiderivative of e^x is:

A. xe^x B. e^x

C. $\ln x$ D. $1/e^x$

MCQ B · integration by parts and bounds

After S66.

For each question, only one option is correct. Justify your choice orally before consult the solutions.

6. For integration by parts, you must know:

A. two functions of class $\mathcal{C}^1$ on $[a;b]$ B. a geometric sequence

C. a fixed point D. a binomial distribution

7. $\int_a^b u'v\,\mathrm{d}x$ is equal to:

A. $[uv]_a^b - \int_a^b uv'\,\mathrm{d}x$ B. $\int_a^b uv\,\mathrm{d}x$

C. $[u+v]_a^b$ D. 0

8. If f is increasing and positive on $[0;1]$, left-endpoint rectangles give:

A. an upper bound B. a lower bound

C. always the exact integral D. a negative value

9. A sequence of integrals $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, f continuous and positive on $[0;1]$, may often be bounded using:

A. $0 \le x^n \le 1$ B. $x^n = n$

C. $\ln x \ge 0$ D. the derivative of n

10. A unit of the integral of a flow rate (L/min) over a time in min is:

A. $\mathrm{L/min^2}$ B. L

C. $\mathrm{min/L}$ D. without a unit

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. An integral is always positive.

2. Two antiderivatives of the same function on an interval differ by a constant.

3. A rectangle approximation removes the need to check the direction of the bounds.

Review of prior work

I can interpret sign, area and mean value; choose an antiderivative or integration by parts, and bound.

To watch out for: confusing geometric area with algebraic integral; forgetting the bounds in $[uv]_a^b$.